

Independent effects of transcranial direct current stimulation and social influence on pain

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Abstract

Transcranial direct current stimulation (tDCS) is a noninvasive neuromodulatory technique with the potential to provide pain relief. However, tDCS effects on pain are variable across existing studies, possibly related to differences in stimulation protocols and expectancy effects. We investigated the independent and joint effects of contralateral motor cortex tDCS (anodal vs cathodal) and socially induced expectations (analgesia vs hyperalgesia) about tDCS on thermal pain. We employed a double-blind, randomized 2 × 2 factorial cross-over design, with 5 sessions per participant on separate days. After calibration in Session 1, Sessions 2 to 5 crossed anodal or cathodal tDCS (20 minutes 2 mA) with socially induced analgesic or hyperalgesic expectations, with 6 to 7 days between the sessions. The social manipulation involved videos of previous “participants” (confederates) describing tDCS as inducing a low-pain state (“analgesic expectancy”) or hypersensitivity to sensation (“hyperalgesic expectancy”). Anodal tDCS reduced pain compared with cathodal stimulation ($F(1,19.9) = 19.53, P < 0.001$, Cohen $d = 0.86$) and analgesic expectancy reduced pain compared with hyperalgesic expectancy ($F(1,19.8) = 5.62, P = 0.027$, Cohen $d = 0.56$). There was no significant interaction between tDCS and social expectations. Effects of social suggestions were related to expectations, whereas tDCS effects were unrelated to expectancies. The observed additive effects provide novel evidence that tDCS and socially induced expectations operate through independent processes. They extend clinical tDCS studies by showing tDCS effects on controlled nociceptive pain independent of expectancy effects. In addition, they show that social suggestions about neurostimulation effects can elicit potent placebo effects.

Keywords: Transcranial direct current stimulation (tDCS), Pain-relief, Placebo effects, Analgesic or hyperalgesic expectations, Anodal/cathodal tDCS of left motor cortex, Adaptive pain calibration

1. Introduction

Transcranial direct current stimulation (tDCS) is a noninvasive neuromodulation technique used as a treatment for neurological, psychiatric, and pain conditions due to its effects on brain network dynamics and behaviors. Despite early controversy regarding the efficacy of tDCS,^{43,64} many studies have now shown that tDCS induces sufficient current^{72,92} to have reliable dose-dependent effects on motor function^{30,71,81} and clinical effects on pain,^{4,18,61,107} anxiety and PTSD,^{3,90,99} and depression.^{67,80,82}

Transcranial direct current stimulation is thought to modulate neuroplasticity. It enhances long-term potentiation^{2,41,54,87} and brain BDNF^{21,77,106} over the stimulated region by inducing subthreshold modulation of neuronal resting membrane

potentials with polarity-specific effects.^{12,29,70} Although anodal stimulation is thought to have an excitatory effect through depolarization of membrane potentials, cathodal stimulation has an inhibitory effect through hyperpolarization.^{1,58} Transcranial direct current stimulation effects appear to be more spatially specific than initially thought, and high-definition tDCS promises to further increase resolution.^{28,94}

Pain is a particularly important target in both research and clinical applications, and tDCS affects both experimental^{5,14,59} and clinical pain.^{44,61,76} Primary motor cortex (M1) is the primary target for analgesia, and DLPFC is the primary target for modulating emotional and contextual aspects of pain. Both M1 and DLPFC stimulation may activate endogenous opioid systems important for controlling pain.^{33,34} Recent studies using high-definition tDCS on M1 and DLPFC show promise for enhanced efficacy because of its more focal stimulation of targeted regions.^{19,27,28,94}

Although tDCS shows promise in modulating pain, its efficacy varies across individuals and studies. One important source of variability is expectancy effects, which have been largely neglected in previous studies. Expectations can have powerful influences on pain.^{7,26,40,85} The process of receiving tDCS may elicit strong treatment expectations,^{9,46,83} creating placebo effects that could mask the effects of stimulation.^{78,79,97} Sham and M1 tDCS have elicited μ -opioid activation in the periaqueductal gray and precuneus, suggesting that placebo-related factors and M1 tDCS influence overlapping targets.³³ Importantly, placebo and tDCS effects may either be additive or may interact. Additive effects imply that placebo factors and neurostimulation can be combined without interference.^{10,17}

Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

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Interactions imply that the right psychological context could either (1) boost stimulation effects⁸⁴ or (2) mask tDCS effects, so that placebo effects should be minimized to reveal active tDCS effects. However, prior studies have not included the necessary experimental conditions to test whether placebo and active tDCS have additive or interactive effects on pain.

In this double-blind, randomized, crossover study, we independently manipulated treatment expectations (analgesia vs hyperalgesia) and active tDCS treatment (anodal vs cathodal M1 tDCS) in a 2 × 2 design to assess the effects of both treatments and their potential interaction on experimental thermal pain. The expectancy manipulation involved suggestions delivered through videos of confederates describing their experience of tDCS in the same context, providing rich and plausible social information that mirrors the types of interpersonal suggestions common in clinical settings.^{85,86} We hypothesized that both M1 tDCS and social suggestions would influence pain and tested whether their effects were additive or interacted.

2. Materials and methods

2.1. Participants

Thirty-one healthy participants were recruited from Dartmouth College and the surrounding New Hampshire community (15 women, $M_{\text{age}} = 24.6 \pm 4.4$ years). Exclusion criteria included age <18 or >55 years, chronic pain, current or history of major psychiatric or neurological diagnosis, substance abuse in the past 6 months, or having contraindications for brain stimulation or induced physical pain (eg, pregnancy, claustrophobia). Participants were also excluded if they were part of the Psychological and Brain Sciences department. The study was approved by the Dartmouth College Institutional Review Board, and informed consent was obtained from all participants before every session. Six participants were excluded based on their calibration results, and 5 participants dropped out of the study during or after the first tDCS session, leaving a final sample size of $n = 20$.

2.2. Adaptive thermal calibration procedure (session 1)

Eligible participants were invited to participate in an initial adaptive thermal calibration procedure. This method can improve the accuracy of the measurement because it adjusts the temperature based on the participants' reported sensation and allows for the determination of an individual's pain and tolerance levels. Four runs (6 trials in each run and 24 trials totally) of noxious heat were administered to 6 forearm sites using iterative adaptive logistic regression. Each stimulus lasted 13 seconds, with 1.5 seconds for ramping. Sensation was rated on a circular scale from "no sensation" to "strongest imaginable." Participants reported pain and tolerance with "yes" or "no" responses to "Painful?" and "Tolerate?" questions. The first calibration run familiarized participants with 47°C heat on all sites. The second run included 45°C, 47°C, and 48.9°C stimuli twice, randomly distributed. Runs 3 and 4 included low and high temperatures based on the pain and tolerance responses of previous trials. A logistic regression based on a 90% response to pain and tolerance of previous trials was used to determine the low and high threshold and was updated after each trial.

After the final trial, low (pain) and high (tolerance) temperatures were determined using logistic regression across all trials, and 2 sites with the least mean absolute error values in rate-temperature linear regression were chosen for pain stimulation. Participants were excluded if their pain and tolerance difference

was less than 1°C. In total, 31 participants completed the calibration session.

2.3. Experimental sessions (sessions 2-5)

Participants eligible after the calibration session were asked to return for 4 more sessions. Each session followed the following design: 4 eight-minute pain tests, social manipulation, a questionnaire about the social manipulation experience, 20-minutes tDCS, a questionnaire about the tDCS experience (3 minutes), 4 eight-minute pain tests (32 minutes total), and a final questionnaire. Participants were randomly assigned to one experiment order (**Fig. 1**). Treatment order in sessions 2 to 5 was randomized, with 6 to 7 days of interval to minimize carryover effects.

2.4. Heat stimulations

Noxious heat was delivered to 2 locations on the right forearm using a TSA-II Neurosensory Analyzer (Medoc Ltd., Ramat Yishai, Israel.) with a 16-mm thermode end plate. Participants received 4 runs of eight-minute heat stimulations before and after the social manipulation and tDCS stimulation (8 runs total). Both low and high heat temperatures and the 2 skin site locations were selected based on the participant's calibration session. Each run included 12 trials of 6 low and 6 high temperatures applied randomly. Each trial lasted 40 seconds and started with "Get Ready" message (1 seconds), jitter (4-6 seconds), heat stimulation (13 seconds including 1.5 seconds of ramp up/down, 10-second plateau), jitter (5-7 seconds), pain rating (7 seconds), and jitter (6-10 seconds).

2.5. Social manipulation

Following the pain tests, participants were told that they would be receiving 1 of 2 possible tDCS protocols made by Defense Advanced Research Projects Agency (DARPA). It was explained that each protocol is based on effects established in the scientific literature, and the protocols have distinct effects on bodily sensation, cognition, and memory and would be delivered in separate sessions on different days. Participants understood that one stimulation type relieved pain and depression in military personnel who had experienced injury or trauma. This same type also decreased pain sensations and pain unpleasantness and improved mood and concentration in healthy individuals. The other stimulation type was described as being useful for recovering loss of feeling and increasing bodily sensitivity in military personnel who had experienced nerve injury and caused increased sensations of pain and touch and increased unpleasantness of bodily sensations in healthy individuals. The cover story was designed to provide a plausible reason why both analgesic and hyperalgesic stimulation would be provided. After these instructions, participants watched a brief video of a previous participant (actually a confederate) describing their experiences with the same tDCS protocol they would be receiving. The confederate described the analgesic stimulation as "floating" and "disconnected from my body," with pleasant sensations. The confederate described the hyperalgesic stimulation as "pinching" and "shocking," with unpleasant sensations. There were 2 male and 2 female confederates, with a hyperalgesic and analgesic expectancy video for each confederate (8 videos total). Participants were presented either a single hyperalgesic or analgesic video from each of the 4 confederates, randomized across

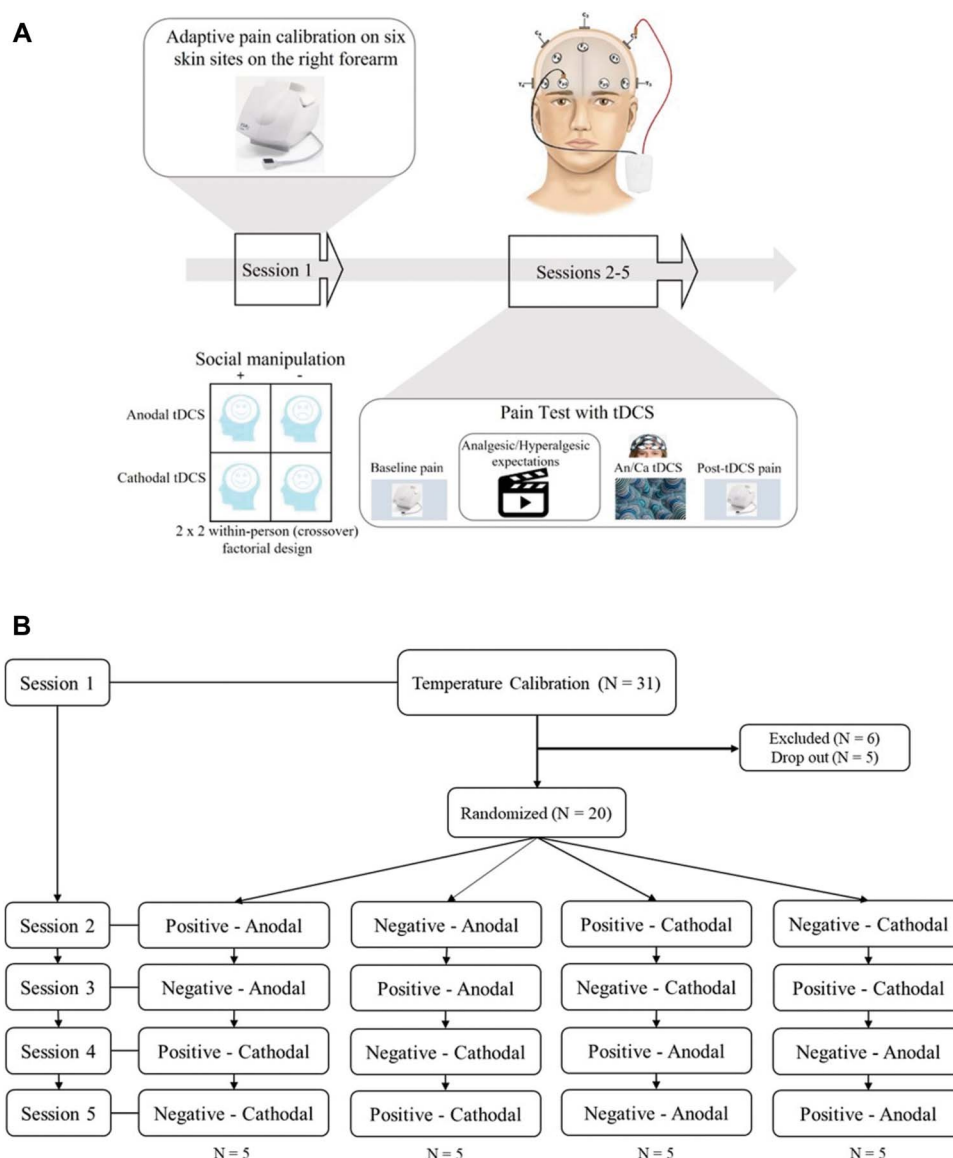


Figure 1. Overview of paradigm and design.

sessions with the constraint that no confederate model appeared in more than one condition. Thus, across participants, the identity and gender of the model was balanced across all 4 conditions.

After the social manipulation phase, participants completed a brief questionnaire with a series of expectation-related questions about their next tDCS and noxious heat sensation experience.

2.6. Transcranial direct current stimulation design

The tDCS device was applied at 2 mA for 20 minutes (with 30-second ramp-up and down to minimize any stimulation discomfort) using the StarStim system (Neuroelectronics, Spain). Electrodes were placed on the C3 (anode) and FP2 (cathode) regions according to the 10-10 EEG system. During stimulation, a random fractal image was displayed on the participant's computer screen. Originally, the main purpose of presenting a random image during each tDCS stimulation was to facilitate a 1-month follow-up session for a memory test planned for a future study. The assignment of pictures to

task conditions was randomized for each participant, and therefore, picture identity was not confounded with task variables.

After tDCS, participants completed brief questionnaires regarding the sensations experienced during tDCS (eg, itching, tingling).⁶ Participants then repeated 4 runs of heat stimulation delivered to the right forearm randomized to the low and high temperatures and the 2 skin locations from calibration. Participants completed a final survey with a series of questions about the participant's experiences during the day of the experiment (eg, regarding alertness, concentration, pain sensation) (Supplementary Materials, <http://links.lww.com/PAIN/C96>).

2.7. Analysis model

The study included a $2 \times 2 \times 2$ (tDCS type $\times$ social manipulation $\times$ stimulus intensity) design. The main planned effects were on the average results across intensities and included the tDCS, social, and tDCS $\times$ social interaction. Our statistical model also assessed effects of interactions with

stimulus intensity. Any trial data (rating) that filled more than 3 standard deviations from the participant's mean value within each of the 8 cells in the $2 \times 2 \times 2$ design was excluded. Less than 1% of trials were excluded based on this criterion. The main statistical model was a mixed-effects model with random intercepts and slopes for tDCS type, social manipulation, and stimulus intensity and their main effects and interactions. For this purpose, full trial-level data were included as well as covariates such as pre-treatment mean pain scores and individual differences in treatment order (anodal first vs cathodal first and analgesic vs hyperalgesic expectancy first). We considered additional covariates for sensations elicited by tDCS type (eg., itching and tingling), but anodal and cathodal tDCS produced equivalent self-reported sensations, so we did not include these covariates. The Satterthwaite method was employed to estimate the degree of freedom. The mixed-effects model used in this study is summarized as follows:

$$\begin{aligned} \text{Pain_post} \sim & 1 + \text{Pain_pre} + \text{Stim_Order} \times \text{Social_Order} \\ & + \text{Days} + \text{Runs} + \text{Trials} + \text{tDCS Type} \\ & \times \text{Social Manipulation} \times \text{Temperature} \quad (1) \\ & + (1 + \text{tDCS Type} \times \text{Social Manipulation} \\ & \times \text{Temperature} \mid \text{Participant}) \end{aligned}$$

In Equation 1, Pain_post is the pain on each trial after tDCS treatment. tDCS Type represents the type of stimulation (anodal or cathodal, coded with 1 and -1, respectively), and social manipulation represents the type of social manipulation (analgesic or hyperalgesic expectancy, coded with 1 and -1). Pain_pre is the mean rating of pain for each temperature before the tDCS and social manipulation in each session. Stim_Order (1 = anodal first, -1 = cathodal first) and Social_Order (1 = analgesia first, -1 = hyperalgesia first) refer to the type of stimulation delivered first for a participant. Days, runs, and trials refer to linear effects of testing day (4 total, mean centered), run number (run within day across 4 runs, mean centered within day), and trial number (trial within run across 12 trials, mean centered within run), respectively. These variables are included to control for habituation and/or sensitization across time. Temperature represents the stimulus intensity used in the calibration (high or low), and participant represents the random effect of the participant.

As we delivered 20 minutes of tDCS but tested effects on pain over a 35-minute poststimulation period, we conducted secondary analyses to assess durability of the tDCS and social

manipulation effects with an extended mixed-effects model. Such effects would be indicated by interactions between experimental manipulations and both runs and trials within run. These interactions were not significant (see Results), so we opted to keep the original model (see above) as the main analysis model. However, all the experimental effects in the main model were also significant in the extended model, so the choice of interactions included does not impact the study findings or conclusions.

2.8. Mediation

We conducted a mediation analysis to investigate the role of pain-expectancy ratings (“analgesic expectancy” vs “hyperalgesic expectancy,” ie, reverse coded pain expectancies) as a mediator of the relationship between the social manipulation and pain sensation after tDCS. The mediation analyses tested whether there was an effect of social manipulation type (analgesic vs hyperalgesic expectancy) on analgesic expectancy ratings (path $a1$), whether analgesic expectations were predictive of pain ratings controlling for social manipulation (path b), and whether the relationship between social manipulation and pain ratings (path $c1$) was mediated by analgesia-expectancy ratings ($c1 - c1' = a1 \times b$). We included tDCS type, temperature, runs, trials, and mean ratings before the tDCS as covariates in the mediation models to control for parameter variation. The significance of all effects were tested using a bias corrected and accelerated bootstrap ($N = 10,000$).^{36,89} In addition, a similar mediation analysis was conducted to investigate the role of pain-expectancy ratings as a mediator variable in the relationship between tDCS type and pain sensation after tDCS (path $a2$, b , $c2$, $c2'$). Mediation analysis was conducted using the CANlab Multilevel Mediation toolbox (canlab.github.io).⁹¹

3. Results

3.1. Calibration

The mean $\pm$ standard deviation values for pain and tolerance were $46.46 \pm 0.76^\circ\text{C}$ and $48.4 \pm 0.6^\circ\text{C}$, respectively. The 6 skin sites on the right forearm, distribution of these temperatures, and the logistic regression for pain and tolerance threshold can be seen in **Figure 2**. This study recruited an equal number of participants from both sexes, and there was no significant difference in the mean values for pain and tolerance between men and women ($P_{\text{low}} = 0.82$ and $P_{\text{high}} = 0.62$).

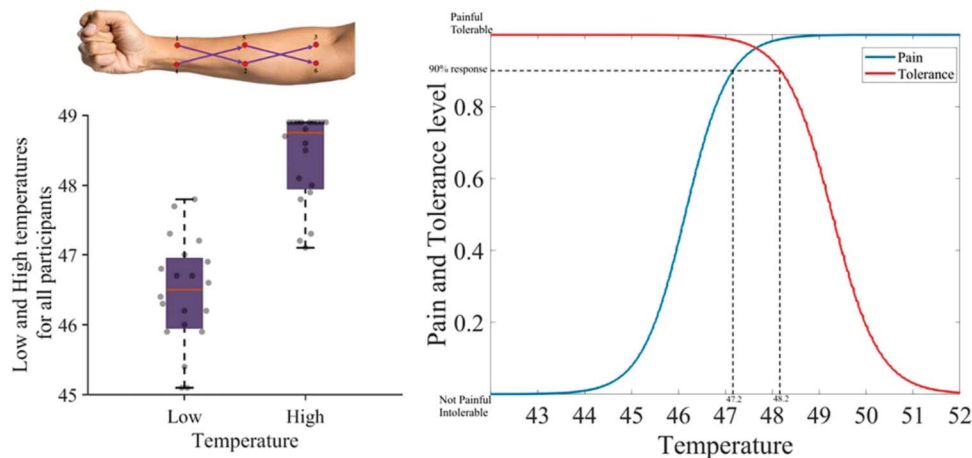


Figure 2. Calibration procedure and results.

3.2. Expectation

As described before, participants watched analgesic (pain-reducing) or hyperalgesic (pain-enhancing) experiences in each session, and after each video, they rated their expectation of experiencing lower or higher pain after that tDCS protocol. The results of the expectation of experiencing lower or higher pain are shown in **Figure 3A and B**. As can be seen in this figure, the expectation of experiencing lower pain after analgesic expectancy-induced videos was significantly higher than that after hyperalgesic expectancy-induced videos ($T(19) = 4.7, P < 0.05$), and the expectation of experiencing higher pain after hyperalgesic expectancy-induced videos was significantly higher than that after analgesic expectancy-induced videos ($T(19) = -4.2, P < 0.05$).

3.3. Sensation after transcranial direct current stimulation

In sessions 2 to 5 after each tDCS stimulation, participants rated the sensations evoked by tDCS. The results for itching and tingling are shown in **Figure 3C and D**. A comparison of the results between anodal and cathodal stimulation sessions showed no significant difference between the sensations evoked by each type of tDCS ($P > 0.05$).

3.4. Effects on pain

As shown in **Figure 4** and **Table 1**, pain was significantly reduced for anodal relative to cathodal stimulation (a main effect of tDCS type, $F(1,19.9) = 19.53, P < 0.001$, Cohen $d = 0.86$). Pain was also significantly reduced for analgesic vs hyperalgesic

expectancy (a main effect of social manipulation, $F(1,19.8) = 5.62, P = 0.027$, Cohen $d = 0.56$). There was no significant tDCS type $\times$ social manipulation interaction ($F(1,19.1) = 1.30, P = 0.267$), consistent with additive effects of both manipulations. As expected, pain was higher with increasing temperature, and this did not interact with tDCS type or social manipulation (see **Table 1** for statistics).

The mixed-effects model controlled for several nuisance variables, including pretreatment pain (Pain_pre), habituation/sensitization (across days, runs, and trials within run), and testing order effects. As expected, posttreatment pain was higher when pretreatment pain was higher on the same day (Pain_pre, $F(1,79.8) = 602.2, P < 0.001$). Pain decreased significantly across days ($F(1,28.7) = 4.37, P = 0.045$) and runs within day ($F(1,3661.1) = 62.02, P < 0.001$) but not significantly across trials within run ($F(1,3663.4) = 3.3, P = 0.069$). We found no main effects of stimulation order (Stim_order; $F(1,18.8) = 0.04, P = 0.842$) or social manipulation order (Social_order; $F(1,19) = 0.12, P = 0.736$) (see **Table 1**).

Figure 4A and B present the posttreatment - pretreatment rating changes ("delta" values) for anodal and cathodal stimulation (x-axis) under analgesic (blue) and hyperalgesic (red) social manipulation conditions, broken down by high and low temperature (left and right panels). Two features of the plots suggest effects that could be followed up in larger-scale, more highly powered analyses. First, effects of anodal vs cathodal stimulation were numerically larger with the analgesic social model, suggesting a potential synergistic interaction, particularly with highly painful stimuli (high temperature). Second, separate mixed-effect analysis of high and low applied temperatures revealed that the effects were

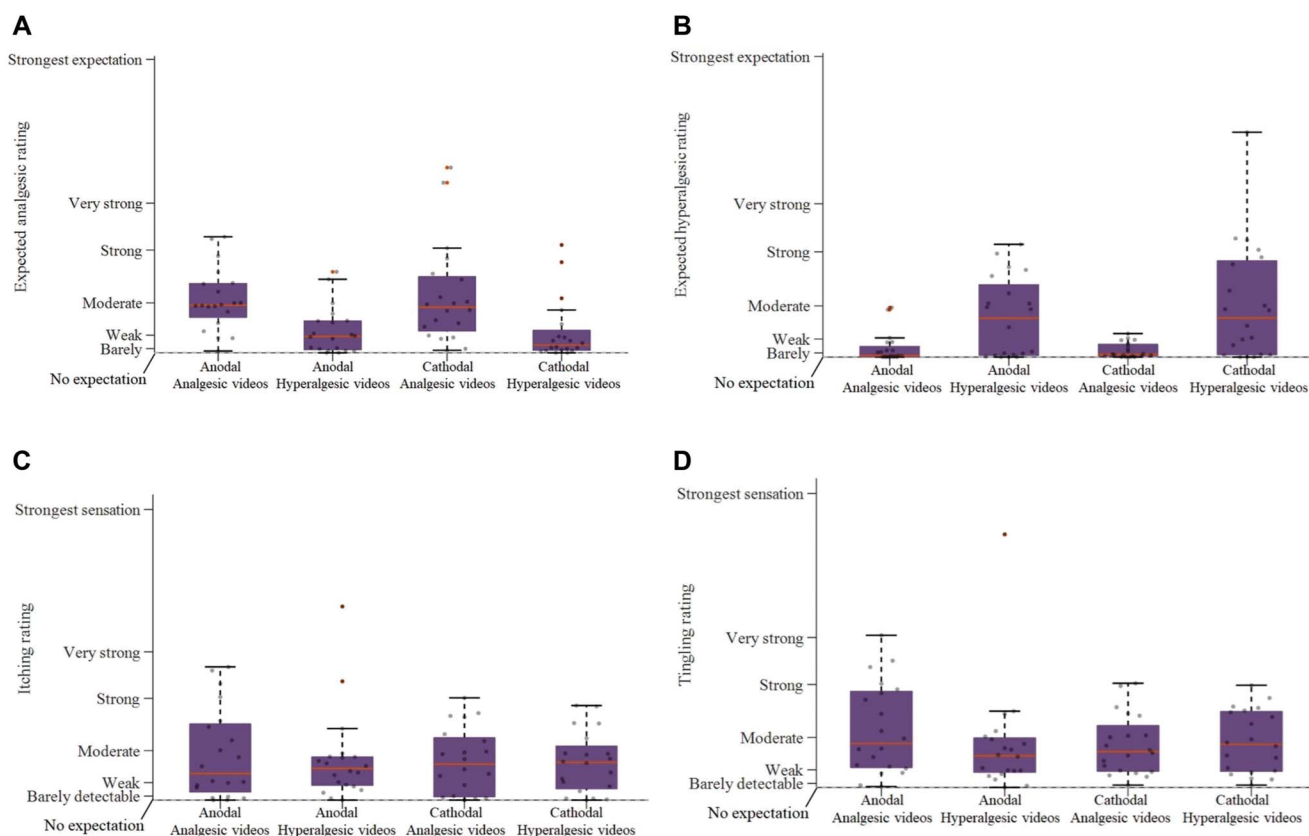


Figure 3. Expectations and tDCS sensations. (A and B) The expectation of experiencing lower pain after analgesic expectancy-induced videos was significantly higher than that after hyperalgesic expectancy-induced videos ($P < 0.05$), and the expectation of experiencing higher pain after hyperalgesic expectancy-induced videos was significantly higher than that after analgesic expectancy-induced videos ($P < 0.05$). (C and D) No significant difference between the sensations evoked by each type of tDCS ($P > 0.05$). tDCS, transcranial direct current stimulation.

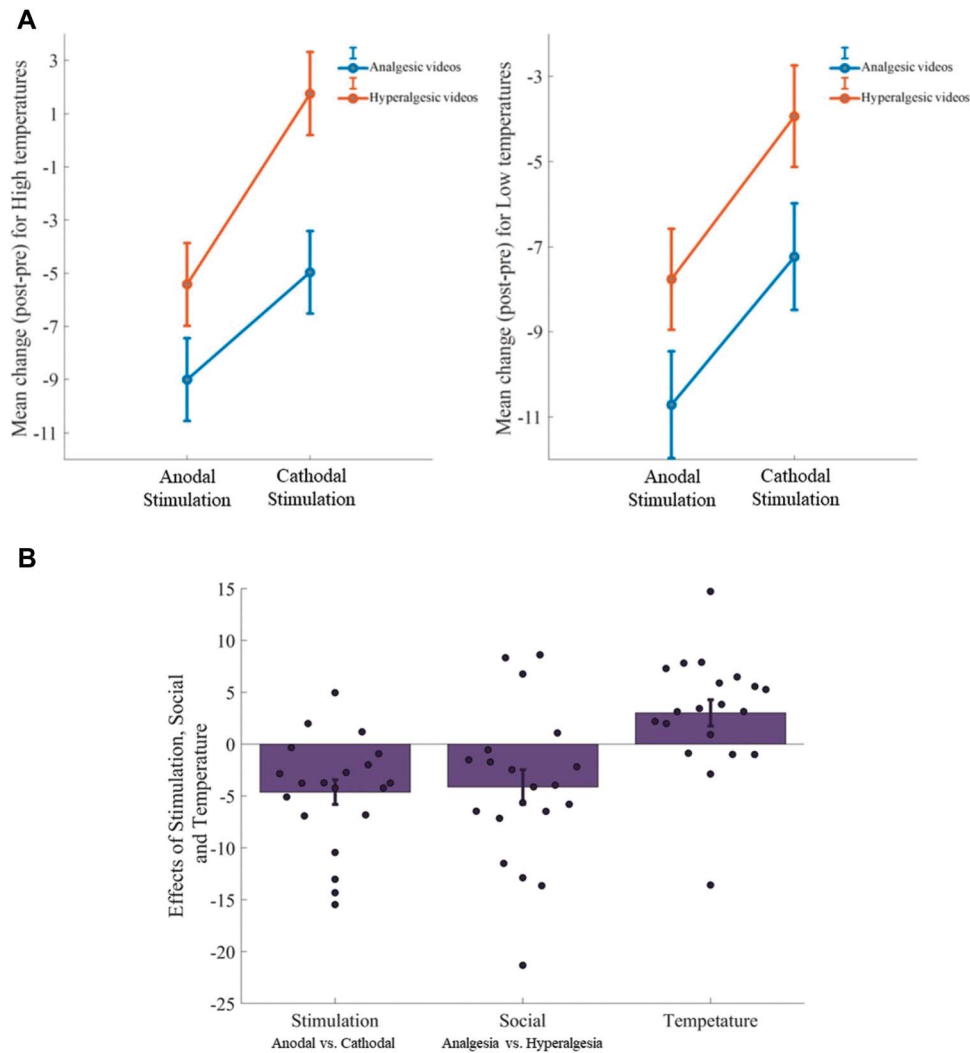


Figure 4. Main findings. (A) Treatment effects (posttreatment – pretreatment change or “delta” values) for each of the 4 experimental conditions and for each of high-temperature (left panel) and low-temperature (right panel) heat. Error bars reflect standard errors of the mean (SEM) after adjusting for random effects of participant.⁷³ (B) The effects of stimulation (anodal vs cathodal), social manipulation (analgesia vs hyperalgesia), and temperature (high vs low). Dots reflect the effect magnitude estimates for individual participants. Error bars reflect the SEM.

numerically larger for anodal vs cathodal stimulation under high temperature and for analgesic vs hyperalgesic social expectancy on low temperature. However, the interactions were nonsignificant, so we do not further interpret these patterns.

We also estimated an extended mixed effects model designed to examine whether the tDCS and social manipulation effects “wore off” over time, which would be indicated by interactions between experimental manipulations and both runs and trials within run. There was no tDCS type $\times$ runs interaction ($F(1,3662.7) = 0.03, P = 0.87$), and a marginal but nonsignificant trend towards reduced social manipulation effects across runs ($F(1,3662.3) = 3.62, P = 0.057$). There was neither a tDCS type $\times$ trials interaction ($F(1,3666.3) = 0.45, P = 0.50$) nor a social manipulation $\times$ trials interaction ($F(1,3665.9) = 0.001, P = 0.97$). The effects of 20-minute tDCS thus appear to be stable across 35 minutes of post-tDCS pain testing. The social manipulation effects also appear to persist across the test but may decrease to some degree across time.

3.5. Mediation analysis

The mediation analysis results, as presented in **Table 2** and **Figure 5**, indicate a significant effect of social manipulation on

expectancy, with analgesic models producing higher analgesic expectancy than hyperalgesic models (Path $a1$, bootstrap $P < 0.001$). Higher analgesic expectancies were associated with lower pain ratings, controlling for both social manipulation and tDCS type (path b , $P < 0.05$). However, the mediation effect was not significant ($a1 \times b$, $P > 0.05$). Moreover, tDCS type had no significant impact on pain expectations (Path $a2$, $P > 0.05$), and expectations did not mediate the effects of tDCS type on pain ($a2 \times b$, $P > 0.05$). Both social manipulation and tDCS type had direct effects on pain independent of expectations, with analgesic social models and anodal tDCS producing lower pain (Path $c1$, $P < 0.01$ and Path $c2$, $P < 0.01$, respectively). **Figure 5A** shows the path diagram for the mediation analysis, with * indicating significant paths and $c1'$ and $c2'$ indicating direct effects of social manipulation and tDCS type, respectively.

As we predicted that expectations would mediate social manipulation effects on pain, and the component effects ($a1$ and b) were both individually significant, we conducted an additional post hoc analysis exploring this effect. Because expectations were only reported once per session at the beginning of the session, we reasoned that they may have a stronger relationship with the social manipulation type and pain

Table 1

Mixed-effects model results in model with random intercepts and slopes for transcranial direct current stimulation type, social manipulation condition, and temperature and the main effects and interactions are reported.

Variable	Estimate	F-stat	DF1	DF2	P
Intercept	1.74	0.61	1	63.7	0.436
Temperature***	3.55	19.74	1	38.8	<0.001
tDCS type (anodal vs cathodal)***	−2.43	19.53	1	19.9	<0.001
Social manipulation (analgesia vs hyperalgesia)*	−1.96	5.62	1	19.8	0.027
Pain pre-tDCS***	0.86	602.20	1	79.8	<0.001
Days*	−0.50	4.37	1	28.7	0.045
Runs***	−0.80	62.02	1	3661.1	<0.001
Trials	−0.06	3.30	1	3663.4	0.069
Stim_order	−0.18	0.04	1	18.8	0.842
Social_order	0.30	0.12	1	19	0.736
Temperature × tDCS type	−0.43	1.92	1	20.7	0.180
Temperature × social manipulation	−0.56	1.59	1	20	0.221
tDCS type × social manipulation	0.58	1.30	1	19.1	0.267
Stim_order × social_order	1.42	2.51	1	19	0.129
Temperature × tDCS type × social manipulation	0.36	0.88	1	20.1	0.359

*** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$ two tailed.

tDCS, transcranial direct current stimulation.

during the early runs in each session. We performed the mediation analysis using only the first 2 runs. In this analysis, expectations significantly mediated the effects of social manipulation ($a1 \times b$, $P < 0.05$) but not tDCS ($a2 \times b$, $P > 0.05$).

4. Discussion

This study aimed to investigate the effects of 2 experimental manipulations—treatment expectations (analgesia vs hyperalgesia) and tDCS (anodal vs cathodal tDCS of M1)—and their potential interaction on experimental thermal pain using a 2×2 design. The effects of tDCS on pain have varied across prior studies, and one source of variability is expectation, which has not systematically been controlled and measured in the literature. In some studies, unblinding of verum vs sham tDCS (which can be perceptibly different) could cause expectancies and tDCS effects to be confounded, making it unclear whether stimulation effects

are caused by the stimulation itself or placebo effects.^{9,33} This is further complicated by the fact that tDCS and other neuro-stimulation treatments may evoke particularly strong treatment expectations.^{9,46,83} This study was designed to both control expectancies and assess whether they mediated tDCS effects. Two key design elements here were (1) the comparison of anodal and cathodal tDCS, which induced equivalent sensations and treatment expectations, and (2) independent manipulations of tDCS and expectations about tDCS effects. Manipulating the latter was intended to both make expectancies more uniform and allow assessment of expectancy-stimulation interactions. In addition, the expectancy manipulation involved a plausible cover story about 2 distinct effects of 2 kinds of stimulation, one conferring hyposensitivity and the other hypersensitivity to the body means of eliciting placebo effects in past research. In some studies, video-based social models as we used here are as effective as live human models.^{8,23} Before each stimulation type,

Table 2

Mediation analysis coefficients and statistics for expectation as a mediator between social manipulation/transcranial direct current stimulation and pain responses.

	a1 Social→ expectancy	b Expectancy→pain	c1' Social→pain (control mediation)	c1 Social→pain (total)	a1b Social mediated by expectancy	a2 tDCS→expectancy	c2' tDCS→pain (direct effect)	c2 tDCS→ pain (total)
Coeff	19.65***	−0.15*	0.07	−2.35**	−2.08	1.90	−2.10*	−1.81**
STE	4.52	0.08	1.91	0.72	1.44	3.04	1.06	0.65
T	4.35	−2.02	0.04	−3.27	−1.44	0.62	−1.99	−2.80
Z	4.83	−2.09	0.03	−3.16	−1.34	0.63	−2.04	−2.80
P	1.34E-6	0.036	0.975	0.001	0.18	0.529	0.041	0.005

See Figure 5 for a graphical description of the model. a1: social→expectancy (social manipulation effects on analgesic expectancy). b: expectancy→pain (analgesic expectancy effects on pain controlling for social manipulation). $a1 \times b$: Mediation of social effects on pain by analgesic expectancy. c1: social→pain (the total effect of social manipulation on pain). c1': social→pain (the direct effect of social manipulation on pain controlling for the mediator (analgesic expectancy)). a2: tDCS→expectancy (tDCS (anodal and cathodal) effects on expectancy). b: expectancy→pain (analgesic expectancy effects on pain controlling for stimulation type (anodal and cathodal)). $a2 \times b$: mediation of stimulation type effects on pain by expectancy. c2: tDCS (anodal and cathodal)→pain (the total effect of stimulation type on pain). c2': tDCS (anodal and cathodal)→pain (the direct effect of tDCS on pain controlling for the mediator (expectation)).

*** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$.

STE, standard error; tDCS, transcranial direct current stimulation.

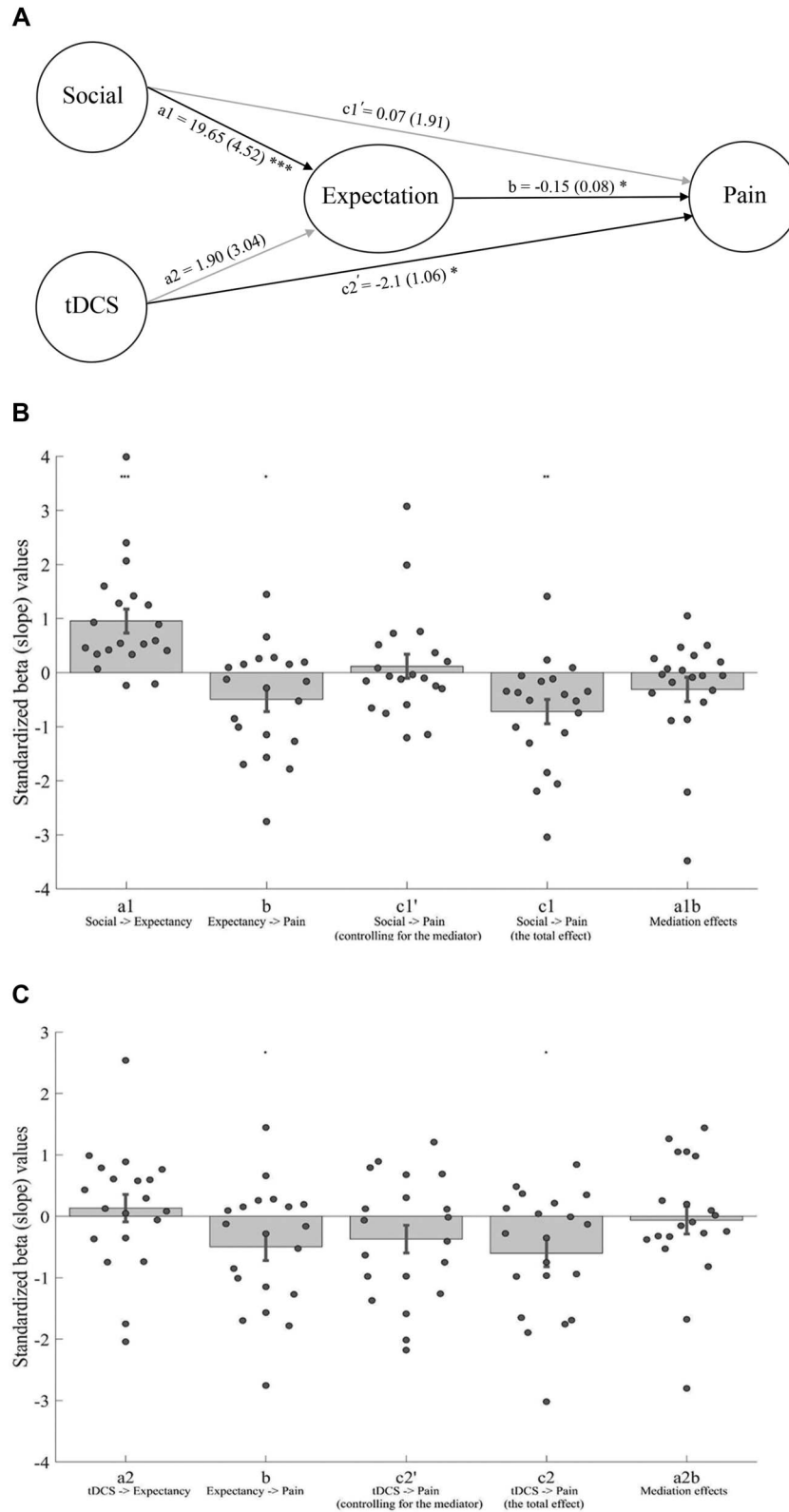


Figure 5. Mediation results. Path a1: social→expectancy (social manipulation effects on analgesic expectancy). (b) Expectancy→pain (analgesic expectancy effects on pain controlling for social manipulation). c1': Social→pain (the direct effect of social manipulation on pain controlling for the mediator (analgesic expectancy)). c1: Social→pain (the total effect of social manipulation on pain). a1b: Mediation of social effects on pain by analgesic expectancy. a2: tDCS→expectancy (tDCS (anodal and cathodal) effects on expectancy). b: Expectancy→pain (analgesic expectancy effects on pain controlling for stimulation type (anodal and cathodal)). c2': tDCS (anodal and cathodal)→pain (the direct effect of tDCS on pain controlling for the mediator (expectation)). c2: tDCS (anodal and cathodal)→pain (the total effect of stimulation type on pain). a2b: Mediation of stimulation type effects on pain by expectancy. tDCS, transcranial direct current stimulation.

participants watched a brief video of a previous participant (actually a confederate) describing their experiences with the same tDCS protocol they would be receiving.

Anodal M1 stimulation reduced pain compared with cathodal stimulation with a large effect size (Cohen $d = 0.86$). Social manipulation of treatment expectancies by presenting videos of confederates posing as participants also had a significant, moderately sized effect (analgesic vs hyperalgesic, $d = 0.56$) comparable to many previous placebo studies.^{22,23,25,39,108,109} Separate mixed-effect analysis of high and low applied temperatures revealed a numerically greater effect of anodal stimulation at high temperatures than low temperatures and a numerically greater effect of social manipulation at low temperatures than high temperatures. This is similar to previous research that has reported greater analgesic effects of tDCS at high levels of pain.⁹ This study revealed that 2 mA tDCS significantly reduced pain sensation, in contrast to a previous study that employed 1 mA anodal and cathodal stimulation with no significant changes in pain sensation or threshold reported.^{45,50,52} Although this demonstrates the potential of 2 mA tDCS to effectively reduce pain sensation, with greater results from anodal stimulation compared with cathodal stimulation, the mechanisms by which tDCS modulates pain-regulating cerebral structures may play a role in this effect. The brain circuits activated in response to strong pain stimulation may be more susceptible to the effects of tDCS at higher temperatures, leading to a greater reduction in pain perception.^{9,42} In addition, the mediation analysis on the first 2 runs of each session showed that expectations significantly mediated the effects of social model but not tDCS.

The effects of tDCS and the social manipulation were additive, with no significant interaction. This suggests that they arise from parallel processes originating in different brain pathways. Prior work in neuroscience is consistent with this notion. M1 stimulation affects pathways connecting motor regions to prefrontal cortex, periaqueductal gray (PAG), thalamocortical connections, and cingulate gyrus.^{9,47,51,65,68,74,98} In addition, descending pathways from the sensorimotor cortex project directly to the spinal cord⁶⁰ and indirectly to sympathetic autonomic neurons in the adrenal medulla.³⁵ These pathways offer a potential biological basis for how M1 stimulation can influence pain-related signaling. By contrast, placebo effects are thought to be mediated by the prefrontal and ventromedial prefrontal cortex (vmPFC), with primary influences on motivational systems. Although some studies suggest that placebo effects may converge with stimulation effects in the PAG and related descending rubrospinal pathways,^{33,38,93,100,101} both early studies^{10,102} and recent meta-analyses^{108,109} suggest that placebo interventions primarily exert their effects through direct influences of cortical (eg, prefrontal cortex) and striatal circuits on pain construction and evaluation. For example, placebo effects and remifentanyl independently contribute to pain reduction, acting through separate neural pathways—remifentanyl targets nociceptive regions like the dorsal posterior insula, whereas placebos affect the lateral prefrontal cortex, converging in the anterior cingulate cortex.^{10,75} In other work, effects of expectation and perceived control on the neurologic pain signature (NPS), a neuromarker related to nociceptive pain, are very small or nonexistent across multiple independent studies¹⁰³ and in a meta-analysis across 20 placebo studies ($N = 603$).¹⁰⁸ A second neuromarker related to pain evaluation, the stimulus intensity-independent pain signature (SIIPS-1), does mediate effects of multiple expectancy manipulations independent of the NPS.¹⁰³ These findings suggest a complex interplay between cognitive expectations and direct analgesic actions, further

complicated by the influence of social factors and expectations on opioids⁶³ and potentially other neurochemical systems. Larger studies are needed to definitively rule out interactions between tDCS and placebo factors and explore potential interactions with different kinds of tDCS and placebo manipulations.

There are some limitations and important future directions to note. In this study, we examined the effect of tDCS on responses to experimentally evoked pain, which offers precise control of nociceptive input and pain intensity. This study focuses on evoked pain in healthy individuals, which is relevant to hypersensitivity to noxious stimuli in chronic pain. However, clinical pain disorders involve much more than nociceptive sensitivity, including hypersensitivity induced by central sensitization in multiple spinal and brain circuits,^{53,56,57,104,105} changes in ongoing clinical pain,⁶² body image,¹⁵ avoidance of movements,^{49,66} social withdrawal,^{11,48} and other depression-related changes.^{24,32,88} Although this study does not capture these many facets of clinical pain, it contributes to a foundation of understanding neuromodulation of nociceptive pain in healthy individuals, which is an important component of clinical pain. Our results provide a normative basis for, eg, identifying pathological alterations in descending inhibition in various chronic pain groups. In addition, this study is a longitudinal, within-person, crossover design, with fewer participants but much more data per participant than standard parallel groups trials. Standard power analyses show that for a given effect size (eg, of tDCS or social placebo), the within-person crossover requires approximately 4x fewer participants for 80% power. Thus, a parallel groups design would need approximately 80 participants to be equivalent in power to this study.

Previous research has targeted various brain stimulation regions, such as the primary somatosensory cortex,^{5,45} primary motor cortex,^{9,47,68} and dorsolateral prefrontal cortex.^{37,96} However, studies that have directly compared M1 tDCS with DLPFC and primary somatosensory cortex (S1) have found that M1 tDCS has a greater effect on pain relief than DLPFC and S1.^{12,51,98} The DLPFC plays an important role in pain perception, notably through its involvement in cognitive and emotional aspects of pain processing. Recent studies have shown the effectiveness of DLPFC tDCS in modulating pain perception,^{13,20,31,69} not only by altering neural activity directly related to pain processing but also by influencing the expectation and placebo effects associated with pain.^{16,55,95,96} Therefore, by modulating DLPFC activity, tDCS could alter the expectancy-related modulation of pain, thereby providing a dual mechanism of action: direct modulation of pain pathways and alteration of pain experience through expectancy mechanisms.

5. Conclusion

In this study, we examined the impact of tDCS, social manipulation, and their interactions on pain relief. Unlike previous tDCS studies in pain relief, this novel research considered the role of expectation in response to tDCS. Despite previous studies considering the effect of tDCS only on pain thresholding, in this study, we investigated the effect of tDCS on pain sensation in several heat trials. Using videos from others' experience, we manipulated participants' expectations towards the effectiveness or ineffectiveness of tDCS. We used an adaptive pain calibration process to individualize heat stimulation and decrease variability in pain response, then participants underwent 4 sessions with different tDCS (anodal and cathodal) and social manipulation (analgesic and hyperalgesic expectancy). Results from a mixed-

effect model analysis showed that anodal M1 stimulation resulted in lower reported pain compared with cathodal stimulation, and analgesic expectancy social manipulation was more effective than hyperalgesic expectancy social manipulation. Future studies may seek to examine whether DLPFC tDCS modulates the effects of expectation on pain perception, in addition to influencing nociceptive pathways, to explore potential dual mechanisms of action. Functional magnetic resonance imaging also holds promise for deepening our understanding of the neural mechanisms underlying pain relief in M1 and DLPFC tDCS.

Conflict of interest statement

The authors have no conflict of interest to declare.

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Data availability: All data used for analyses in this study is publicly available in CANLAB repository at https://github.com/canlab/stimMAP_2022_public.

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